

Adaptive Speaking Style Did Not Hold Up

Geist Atlas · Module 13 · What the Tutor Needs to Know

Liam Magee*

Education Policy, Organization and Leadership
University of Illinois Urbana-Champaign

September 2026

Adaptive Speaking Style Did Not Hold Up

[**SCOPE-BOUND**] Supported within a narrow scope · **Family:** What the Tutor Needs to Know

Claim: A pre-registered two-family test did not find a reliable benefit from adapting speaking style to the learner profile. Terra failed the profile-manipulation gate; Sonnet passed that gate but returned no interaction. A late contrast then froze the teaching entirely — the same scripted moves delivered sarcastically against warmly — and the two deliveries were indistinguishable on the proof-DAG in all three learner characters, an operator-accepted descriptive null.

A 120-dialogue confirmatory study in one simulated detective world. It closes this field selector on the present evidence, while keeping the result separate from any claim about an isolated tutor model or human learning.

Derived view of `paper-full-2.0.md` v3.0.311 — §6.17, §6.29. The paper is canonical; edit it there, not here.

6.17 Register Selection Does Not Produce a Confirmed Profile-Contingent Benefit Across Two Model Families (Pre-Registered Gate)

The register-policy line asked whether a tutor should change its speaking stance according to a learner profile, rather than whether one register is best in aggregate. A frozen confirmatory design crossed three policies (`bland`, the adaptive `field` selector, and a fixed `negative` stress register) with four v3 simulated-learner contracts (`diligent`, `affective_resistant`, `false_memory`, `proof_skipper`) at $n = 5$ per cell on `world_005_marrick`, once on an all-Terra stack

*This sentence is the only one actually authored by Liam Magee in this paper.

and once on an all-Sonnet-5 stack: 60 dialogues per family, 120 total. The primary endpoint was mechanically computed proof-DAG coverage at learner turn 16. Each family also had to pass its own in-run profile-discrimination gate — average pairwise cosine < 0.85 and maximum similarity to `diligent` < 0.90 — before any policy-by-profile interaction could confirm. A block that failed this manipulation gate was instrument-invalid for the family claim, regardless of an attractive cell contrast.

Whole stack	Selected traces	Average profile cosine	Maximum similarity to diligent	Frozen manipulation gate	Interaction verdict
Terra (<code>gpt-5.6-terra</code> at all four seams)	60	0.812	0.912	fail	instrument- invalid
Sonnet 5 (Sonnet at all four seams)	60	0.645	0.694	pass	null

Strict result: no family confirmation; no two-family claim. Terra’s only supported policy-by-profile contrast was produced by the fixed hostile floor: the **negative** contrast for `affective_resistant` relative to its diligent contrast was -0.334 (95% bootstrap interval $[-0.468, -0.167]$). It ran opposite to the pre-registered beneficial-crossing direction and the family failed the binding manipulation gate because its maximum similarity to diligent was 0.912. It is therefore descriptive evidence of a possible stack-bounded contraindication, not a register-selection positive. Sonnet passed the frozen cosine gate, but every primary interaction interval crossed zero. The adaptive `field` policy confirmed no interaction on either stack. Excluding the four Sonnet rows whose runs ended after their valid turn-16 assessment but before full secondary closure leaves 56 rows and the same null verdict.

The comparison is deliberately read as a whole-stack boundary, not a tutor-model isolation. Tutor, automated learner, classifier, and learner-record extractor changed together between Terra and Sonnet; the coverage aggregate is deterministic, but the record from which it is computed still passes through an LLM extraction seam. The large cross-stack level difference can therefore arise from tutor conduct, learner conduct, extraction calibration, or their interaction. What closes is the general claim that this adaptive selector produces a profile-contingent fixed-horizon benefit on this apparatus. The register palette remains a useful implementation vocabulary, and Terra’s hostile-register collapse remains a safety hypothesis, but neither licenses another selector-rescue block.

Status and caveats (per §5.12.6). Pre-registered and complete at $n = 5$ per cell; one detective world, simulated contract learners, one operational fixed-horizon endpoint, and no human-learning claim. Terra has complete secondary traces for all 60 selected rows; Sonnet has 60 primary-complete rows, 56 with full secondary closure, and four primary-only rows. One instrument boundary discovered after the fact: the Sonnet

family block ran before the safe-mode-v1 claude-CLI context-isolation fix, so its speaking-tutor calls ambiently loaded ~16k tokens of repository context (the codex/Terra path was always isolated). This is a whole-family transport condition, uniform within the block — it cannot manufacture the within-family register contrasts, but it adds one more uncontrolled difference to the already whole-stack Terra-vs-Sonnet comparison above, and Sonnet rows from this block must not be pooled with post-fix runs (check run-header provenance before any cross-run pooling; the §6.18-addendum Step 4 run is entirely post-fix). Reproduction is zero-call and hash-verified: the explicit final-row selection, 5,000-draw bootstrap (seed 20260713), profile gates, QA matrix, archive lineage, and strict verdict live in `scripts/analyze-register-confirmatory-step2.js`, `exports/register-confirmatory-evidence/final/`, and `config/adaptive-tutor-evidence/tutor-stub-reg`. The next experiment changes both delivery channel and intervention grain: point-of-action coaching and compiled constraints are a successor test, not a retry of the selector.

6.29 The Edged Register with the Teaching Frozen: Sarcastic and Warm Deliveries of the Same Scripted Moves Are Indistinguishable on the Proof-DAG, in All Three Learner Characters (Operator-Accepted Descriptive Null; Development-Tier)

The edged-register line has run through this paper as a question that kept outgrowing its instruments. The §6.7 arc established that manner presence is measurable by reading and not by phrase-matching, and left its four August runs stack-bounded; the register router (§6.13.19) put a stance shift on a machine-checked outcome and found the tutor’s conduct measurably changed while the outcome stayed identical, 16/20 grounded in both arms; and a 312-row two-scenario outcome block outside the cell harness measured a primary-endpoint drop under sharp delivery (0.601 pooled over two sharp variants against 0.712 warm) that its 48% power could not resolve, then found its one surviving lead — the sharp-arm learner supplies less backing for its claims — uncertifiable by its own instrument, because every reading passed through a model that could see the sarcasm in its window: nothing in that design separates “the learner argued less” from “the reader marked the argument down” (`notes/2026-08-16-edged-register-calibration-draft.md` §3.10; workplan card `edged-register-outcome-study`, closed scope-bound). This section closes the line by removing both confounds at once. The scripted-core tutor stub decides every teaching move in code — which exhibit to release, what to stage, when to close — and a model only words the decided move in a pinned manner, warm or sarcastic; completion is read off the strict proof-DAG, a machine check no reader’s opinion of the manner can reach. If a sharp voice costs or buys the learner anything this endpoint can see, it must show here, where the voice is the only thing that varies.

Two stages, the second sized on the first’s facts. Stage 1 (signed GO 2026-08-27, eval commit 194653a8) ran warm-only calibration: four learner characters × 12 dialogues on world 005 (Marrick), strict proof-DAG, `codex.gpt-5.6-luna` in the tutor, learner, and learner-analysis seats, a 120-call pool and a 40-turn cap. It taught three things. The pool, not the learner, was the modal stop (35 of 48 rows died on it near turn 37, several with the full fact chain assembled and the close unspoken). The bored character never moves under warm

(coverage 0 in all 12 rows), so it cannot carry a contrast and was dropped by the registered rule. And a blind reader (Sonnet 5, sealed per-turn context, arms and outcomes hidden) hears edge in a warm-pinned voice at a rate ordered exactly by learner resistance — bored 25/381 turns (6.6%), diligent 51/401 (12.7%), proof_skipper 49/328 (14.9%), affective_resistant 80/452 (17.7%) — with zero register-pin violations in 1,562 turns. Stage 2 (GO 2026-08-28; design, endpoints, floors, and ceilings written into the sizing memo and ruled on before launch) contrasted a `fixed_sarcastic` against a `fixed_warm` register policy in three characters $\times$ 12 dialogues per arm, 72 in all, with the call pool raised to 180 so the turn cap, not the budget, becomes the binding stop. Registered primary: DAG coverage at fixed learner turns 8, 12, and 16, plus the area under the coverage curve to turn 16 — read on every row, immune to the stop cause. Secondary: closure rate and grounding turn. Manner check: a blind read was attempted on every fixed-policy tutor turn, with a 0.8 presence floor binding the sharp arm and the warm arm’s leak report-only against the Stage-1 baseline. Spend ceiling 12,960 generation calls plus 3,200 fidelity reads; attended; no resampling after a failure; indeterminate-means-stop; a failed row stands and is reported. One pre-block amendment is disclosed in the memo: the first launch stalled on a frozen 42,000-character prompt-audit ceiling one turn behind where the old call pool used to bind, the operator raised it to 64,000, and the six sealed rows of that root stand as records and take no part in any analysis.

Two instrument defects surfaced mid-arc; the affected data was voided or re-run whole, never patched. First, a free pre-payment `--mock` fidelity pass refused a verdict on the first resistant block: 68 register-pin violations over 2,484 fixed-policy tutor turns, every one in the resistant cell. The raw job traces, written live during the run, carry `"selected_register": "warm"` inside sarcastic-pinned jobs: a release-pacing override rewrote the pinned stance whenever the learner asked for more time (130 occurrences of its reason string in the sarcastic arm), and a second override forced plain three times. Stage 1 could not have seen this — it pinned warm, and the pacing override’s target is also warm, so the rewrite was a no-op on every turn. The deeper threat was policy asymmetry rather than register impurity: the sarcastic arm had in effect run a mildly responsive policy — sharp, but yielding on request — against a flat warm arm, and “sharp but yielding beats uniformly warm” is a reading no fidelity read can separate from “sharp beats warm”. The pin is now terminal (`a8e0d82f`, with a regression test requiring every stance override to carry a fixed-pin exemption, a dynamic-policy scope, or the pressure-probe condition); the block was voided under the registered rule and the cell re-run whole. Second, the tutor’s leak guard — the fail-closed check that stops a dialogue when the tutor drafts a fact the code has not released — read its compound word lists as any-word, so the bare word “die” beside the answer’s name (“die-flaw”, “die marks”) fired the private-conclusion kill with no leak present. All 8 recorded kills across the study were this defect; none showed a real leak. Fixed in `d74969e0` (component lists now require every listed word in one sentence; 26 regression tests pin the 8 killed sentences); the killed dialogues re-ran clean on the fixed code. The final roster is 72 of 72 dialogues, 12 per arm in every cell, no exclusions; the 180-call pool never bound, and the turn cap replaced it as the modal stop.

The endpoint estimates are descriptively null in every cell, at every horizon. Coverage at learner turn 16, sarcastic against warm ($n = 12$ per arm):

Character	sarcastic	warm
affective_resistant	0.250 (sd 0.23)	0.292 (sd 0.24)
diligent	0.486 (sd 0.21)	0.472 (sd 0.19)
proof_skipper	0.444 (sd 0.19)	0.486 (sd 0.22)

The earlier horizons agree (turn 8: 0.056 vs 0.028, 0.125 vs 0.084, 0.097 vs 0.084; turn 12: 0.222 vs 0.250, 0.403 vs 0.431, 0.389 vs 0.361), as does the area under the coverage curve to turn 16 (resistant 0.105 vs 0.113; diligent 0.220 vs 0.218; proof_skipper 0.204 vs 0.200). At turn 16 the absolute arm gaps are at most about one fifth of a pooled standard deviation, and the leading manner flips across horizons and characters. Closure tells the same story from the secondary endpoint: proof_skipper grounds 10/12 under both manners (warm faster, mean grounding turn 25.7 against 28.1), the resistant learner grounds 1/24 (the one closure on warm), and the diligent learner grounds 0/24 — it assembles the full chain and does not speak the close, under either voice. The voided block’s one live effect — sarcastic ahead of warm on the resistant learner, 0.455 against 0.278 at turn 16 — did not survive the clean re-run, where the direction reversed (0.250 against 0.292).

The treatment was delivered; the descriptive null is not explained by observed pin failure. The blind reader attempted all 2,508 fixed-policy tutor turns and classified 2,505. Two proof_skipper warm-arm turns were unread, as was one resistant sharp-arm turn. Sharp-arm manner presence: 0.848 (306/361) in proof_skipper, 0.890 (388/436) in diligent, 0.900 (432/480) in the re-run resistant cell — every arm over the registered 0.8 floor on the recorded denominator. The unread resistant turn is a measurement refusal: the provider’s API refused that turn with a harms label on every retry, because the sealed reader context strips the tutorial framing from a counterfeit-die discussion. The instrument therefore wrote its registered verdict — *incomplete*, no study-level fidelity verdict while any sharp turn is unread. The operator separately accepted the arm for development-tier descriptive interpretation at 479 of 480; that ruling is recorded beside the instrument verdict and does not convert it into a preregistered pass. The register pin held everywhere on the fixed code: zero violations across all 2,508 fixed-policy turns, including 40/40 sarcastic turns in each of the twelve re-run resistant dialogues. The deterministic harm scan was empty over the same 2,508 turns: zero word-list matches, zero confirmed person-attacks. The warm arms leaked edge at 10.7–11.9% of classified turns, report-only, on which more below.

Two findings stand beside the null. First, the leak channel: learner resistance can pull edge into a voice pinned warm. The Stage-1 ordering is exact — 6.6% (bored) to 17.7% (resistant) — but Stage 2 does not reproduce it: its three warm arms read edged at nearly the same rate (proof_skipper 40/337, 11.9%; diligent 51/428, 11.9%; resistant 50/466, 10.7%), so the ordering is a finding about Stage 1’s harness and four-character roster, and the channel’s stability across designs is an open question, reported here with both stages’ denominators so no successor repeats the raw-count reading. Second, the outcome is character-bound, not manner-bound: what a learner character is disposed to do with the proof sets coverage and closure, and how the tutor sounds while making the same moves sets nothing this endpoint can see.

What may be claimed. Claimable: with the teaching fixed by a scripted core, the manner of its delivery — warm against sarcastic — produced no detectable difference in proof progress at any registered horizon, in any of three learner characters, on this stack; and the report-only observation that a blind reader hears edge in warm-pinned tutor turns at rates between 6.6% and 17.7% across stages and characters. Not claimable: that register does not matter for learning — the design froze exactly the channel through which a manner would plausibly act, namely the moves it licenses; that sarcasm is safe or unsafe for learners (the harm scan was empty, but the learners are simulated); anything about human learners, other worlds, other stacks, or manners beyond this warm/sarcastic pair. The reading the arc leaves behind is the one §6.13.19 reached from the other side: register indexes the relationship and changes the tutor’s conduct, and its causal power over outcomes — if it has any — travels through the moves it licenses. A design that lets tone drive the moves would be a new study under its own registration; none is planned.

Status and caveats (per §5.12.6). Pre-registered design: the Stage-1 GO signed 2026-08-27; Stage-2 design, primary endpoints, fidelity floors, and spend ceilings written and ruled on before launch, with amendments and block reports disclosed in dated sections of the sizing memo (`notes/2026-08-28-edged-register-stub-stage2-sizing-memo.md`). The registered fidelity gate did not issue a study-level verdict because one sharp-arm turn remained unread; the endpoint pattern is therefore reported as an operator-accepted, development-tier descriptive null, not a preregistered null result. Two mid-arc instrument repairs (`a8e0d82f, d74969e0`) followed the standing rule — the affected block voided or its rows re-run whole, no measured row patched, no historical row rescored. Development-tier: one world (005, Mar-rick), a scripted-core stub, simulated learners, one model family in all generating seats, one blind reader (`claude-code/claude-sonnet-5`) on the validated manner-presence instrument. Attrition: none on the final dialogue roster (72/72); two learner-analysis classifier calls failed all retries at turn 18 of two diligent dialogues, past the turn-16 endpoint horizon; three of 2,508 fidelity-turn reads remained unread (two warm, one sharp). Attended throughout; no resampling; the re-runs ran on the operator’s word against the registered ceilings. Provenance: final trajectory report `.tutor-stub-auto-eval/stage2-trajectory-report-final.md`; fidelity reports in the three sealed cell roots (`qa-matrix-2026-08-28T23-01-11-203Z`, `qa-matrix-2026-08-29T00-13-58-641Z`, `qa-matrix-2026-08-29T12-21-26-240Z` with their resume chains); both voided roots retained (`qa-matrix-2026-08-28T20-52-23-839Z`, prompt-ceiling stall; `qa-matrix-2026-08-28T21-41-13-718Z`, pin leak); artifacts archived in the private repo (commits `90d43ae6`, `0af1573c`, `95c80b75`); workplan card `edged-register-stub-dag-replication` (closed). Exports-only: no DB writes, no rubric changes, no abstract or headline-N changes; no sections renumbered (§6.29 precedes §7).

Neighbouring modules: Module 7 — A Bigger Learner Model Did Not Help; Module 3 — The Tutor Did Not Become More Adaptive; Module 12 — More Layers Do Not Mean More Benefit; Module 4 — How We Checked Every Claim.